

Organization of a Computer

Lesson 04 — Units, Buses and the Instruction Cycle

JKSSB Computer Awareness

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Video Lesson 04

Watch: How Computers Actually Work: CPU, Memory & I/O
Explained

The Office Desktop as a System

- ▶ The clerk types marks on the **keyboard** and clicks with the **mouse**; the scanner feeds a form.
- ▶ The **CPU** processes the marks using a spreadsheet formula.
- ▶ The **monitor** shows the total and the **printer** prints it.
- ▶ **RAM** (Random Access Memory) holds the work while it runs; the **SSD** keeps the file after the machine is switched off.

One familiar machine carries every unit in this lesson.

Five Functional Units

- ▶ **Input unit**: accepts data and instructions.
- ▶ **Output unit**: presents the result.
- ▶ **Memory unit**: primary storage that holds data and instructions currently in use.
- ▶ **Arithmetic Logic Unit (ALU)**: performs arithmetic and logical operations.
- ▶ **Control unit**: directs the sequence of operations and the flow of data.

High-yield grouping

The ALU and the control unit, together with registers, form the **Central Processing Unit (CPU)**.

Inside the CPU

Arithmetic Logic Unit

- ▶ Adds, subtracts, compares
- ▶ Works on binary data
- ▶ Produces a result and status flags

Control Unit

- ▶ Fetches and decodes instructions
- ▶ Sends control signals
- ▶ Coordinates all other units

Registers are small, fast storage inside the CPU; the program counter and instruction register are examples.

Input and Output Units

Input devices

- ▶ Keyboard, mouse
- ▶ Scanner, microphone
- ▶ Light pen, digitizer

Output devices

- ▶ Monitor, printer
- ▶ Speaker, plotter
- ▶ Sound card output

Classify the device by the role it plays in the described task.

Memory (Primary) versus Storage (Secondary)

Primary storage / main memory

RAM and ROM — directly accessed by the CPU; holds data and instructions currently being processed. RAM is volatile.

Secondary storage

Hard disk, SSD and pen drive — keeps files permanently, but is not accessed directly during processing.

The CPU works with main memory; secondary storage is reached through main memory and the operating system.

The Three Buses

Bus	Direction	Carries
Address bus	CPU → memory	The address of a memory location
Data bus	Bidirectional	Data and instructions between CPU and memory
Control bus	CPU → units	Control and timing signals

Address = where; data = what; control = when and how.

Address vs Data vs Control Bus

- ▶ **Address bus**: one-way, from the CPU to memory — it carries the location being selected, not the contents.
- ▶ **Data bus**: two-way — it moves the actual data and instructions in both directions.
- ▶ **Control bus**: one-way control path — it carries read, write and timing signals.

Near-neighbour trap

The address bus does *not* carry data, and the data bus does *not* carry addresses. Do not swap them.

Motherboard, Slots and Interfaces

- ▶ The **motherboard** is the main circuit board linking the CPU, memory and buses.
- ▶ **Expansion slots** let you add cards such as a graphics card or a network card.
- ▶ **Interfaces/ports** connect external devices such as the keyboard, printer and monitor.
- ▶ The system bus runs along the motherboard between the CPU and the other units.

The Stored-Program (von Neumann) Idea

- ▶ Early machines were rewired for each task.
- ▶ The **stored-program concept** keeps the program's instructions in the same main memory as the data.
- ▶ The CPU fetches an instruction, then processes data under it — no rewiring needed.

Association

The idea that both instructions and data are stored in memory is associated with **von Neumann architecture**.

The Instruction Cycle

1. **Fetch**: the CPU obtains the next instruction from main memory.
2. **Decode**: the control unit works out what the instruction means.
3. **Execute**: the ALU or another unit performs the operation.
4. **Store** (when required): the result is written back to memory or a register.

Order cue

The cycle repeats in the fixed order fetch → decode → execute.

Data and Instruction Flow

- ▶ Input data and instructions enter through input devices.
- ▶ They are held in main memory and fetched by the CPU over the system bus.
- ▶ The CPU decodes and executes, using the ALU and registers.
- ▶ Results go to output devices or back to main memory.
- ▶ Secondary storage keeps files between sessions — the CPU does not reach it directly.

Lesson 04: Recall

- ▶ Five units: input, output, memory, ALU, control unit; CPU = ALU + control unit + registers.
- ▶ Address bus: CPU → memory, carries the address. Data bus: bidirectional, carries data and instructions. Control bus: carries control signals.
- ▶ The CPU directly accesses main memory (primary storage), not secondary storage.
- ▶ Stored-program (von Neumann): instructions and data share memory.
- ▶ Instruction cycle: fetch → decode → execute.